EXPERIMENT-2

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USER INTERFACE AND DESIGN

Mind Vault: A Game-Based Learning System for Chunking, Recall, and Recognition

FRAME 1: INSTRUCTION PAGE

Analysis of the Mind Vault – Instruction (Start Menu) Page

The instruction page of the *Mind Vault* application introduces users to a memory recall task based on numeric data memorization. This screen prepares users for the task by clearly explaining the objective, flow, and rules of the game while applying chunking principles to reduce cognitive overload.

1. Clear and Sequential Information Presentation

- Instructions are displayed in a simple and structured manner.
- Each instruction conveys a single rule or action.
- This helps users quickly understand the task before starting.

2. Logical Grouping of Information

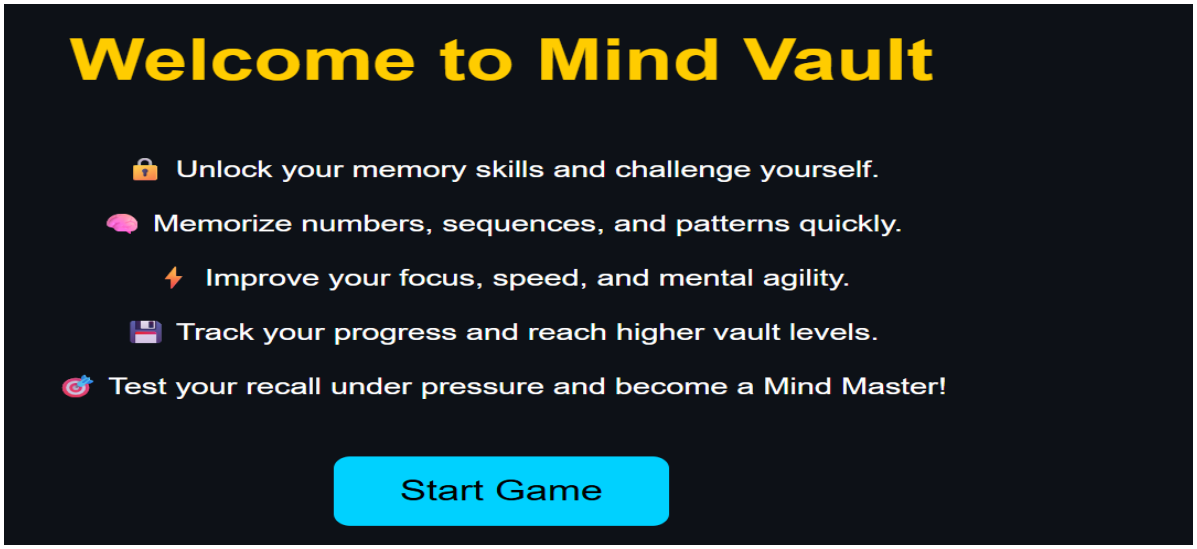
- **Memorization Phase** – Users are informed that a numeric sequence will be displayed.
- **Time Constraint Awareness** – The sequence will be visible only for a limited duration.
- **Recall Phase** – Users must enter the memorized access code.
- **Progression Rule** – Correct recall allows users to advance to higher vault levels.

3. Visual Hierarchy and Design Elements

- The title **"Mind Vault"** is prominently displayed.
- Memory-related icons represent thinking, data storage, and speed.
- The **Start** button is highlighted to indicate the primary action.

4. Simplicity and Clarity

- Short and direct sentences are used.
- Minimal design reduces distractions and cognitive load.
- The instruction page ensures user readiness before beginning the task.



FRAME 2: CHUNKING PHASE

Analysis of the Mind Vault – Chunking Phase Screen

This screen represents the Chunking Phase of the Mind Vault application, where users observe and memorize a numeric data stream within a limited time. This phase focuses on memory encoding using chunking strategies.

1. Purpose of the Screen

- Serves as the memory encoding stage.
- Users visually process and store numeric information.
- Encourages grouping of numbers into meaningful chunks.

2. Key UI Components

- **Countdown Timer** displays the remaining memorization time.
- **Numeric Data Stream** is shown centrally for maximum focus.
- Adequate spacing allows users to apply chunking strategies.
- Status text such as *"Memorizing…"* indicates the current phase.

3. How the Chunking Phase Works

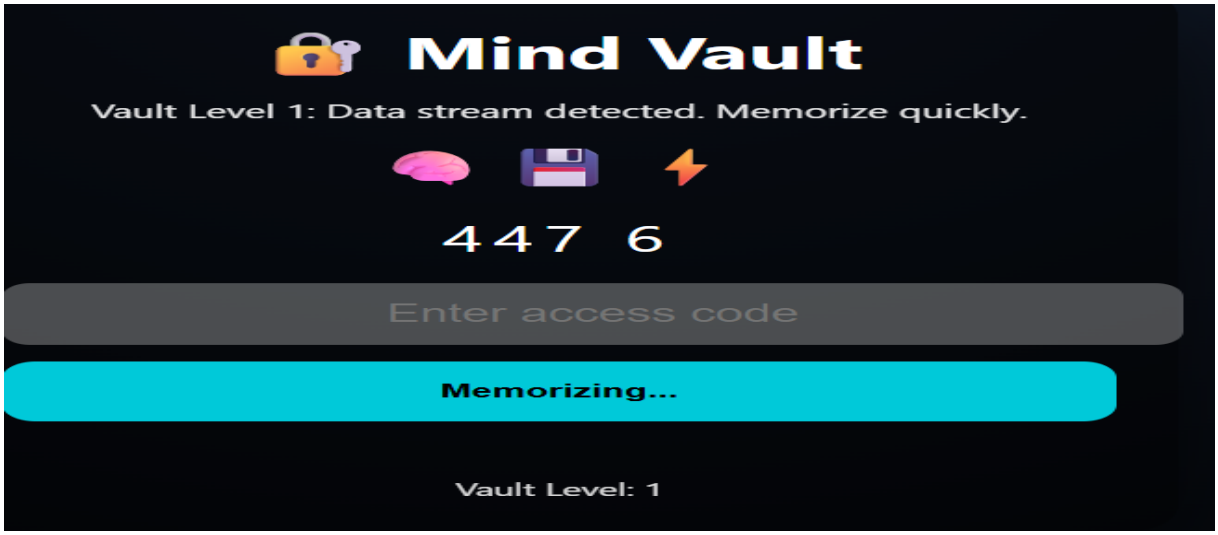
1. Users observe the displayed numeric sequence.
2. They group numbers mentally using chunking techniques.
3. The countdown timer limits observation time.
4. The data disappears once the time ends.
5. Users are transitioned to the recall phase.

4. Cognitive and UX Benefits

- Improves short-term memory retention.

- Reduces cognitive overload.
- Enhances pattern recognition and structured recall.

Mind Valut: Access Granted



FRAME 3: RECALL PHASE

Analysis of the Mind Vault – Recall Phase Screen

The Recall Phase evaluates the user's ability to retrieve and reproduce the memorized numeric data from the Chunking Phase.

1. Purpose of the Screen

- Acts as the memory retrieval stage.
- Measures recall accuracy.
- Tests the effectiveness of chunking strategies.

2. Key UI Components

- Title indicating the Recall Phase.
- Instruction text guiding the user to enter the access code.
- Numeric input field for recalled data entry.
- Submit button to confirm the response.
- Vault level indicator showing difficulty progression.

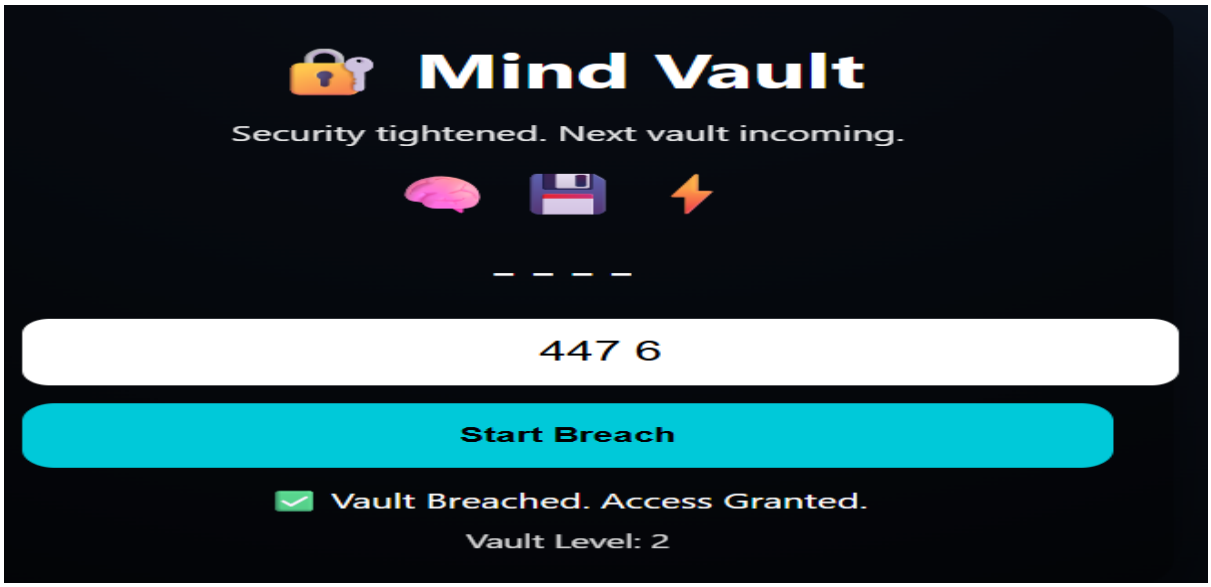
3. How the Recall Phase Works

1. Users retrieve the memorized numeric sequence.
2. They enter the recalled data into the input field.
3. The submit button confirms the response.
4. The system validates the input.

5. Users are redirected to the result screen.

4. Cognitive and UX Benefits

- Encourages active recall.
- Evaluates memory accuracy.
- Simple layout minimizes distraction.



FRAME 4: RESULT PAGE

Analysis of the Mind Vault – Result and Feedback Screen

This screen represents the Result Phase, where users receive feedback based on their recall accuracy.

1. Purpose of the Screen

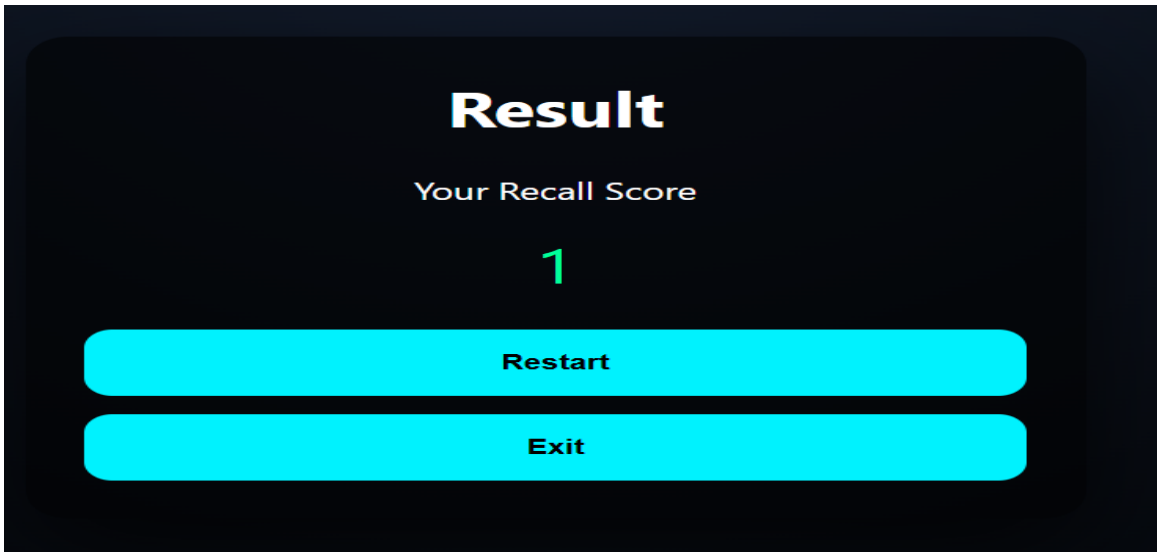
- Displays the user's recall performance.
- Provides feedback on accuracy.
- Allows users to choose the next action.

2. Key UI Components

- Title reinforcing the game identity.
- Score display showing correct recall results.
- Feedback message indicating performance.
- Action buttons:
    - **Continue** – Proceed to the next level.
    - **Restart** – Retry the current task.
    - **Exit** – End the game session.

3. How the Result Phase Works

1. The system compares user input with the original sequence.
2. The recall score is calculated.
3. Results are displayed clearly.
4. Users select their next action.

4. Cognitive and UX Benefits

- Immediate feedback improves learning.
- User control enhances engagement.
- Gamified progression motivates repeated attempts.